Airlytics: Intelligent Flight Fare Estimation & Explainable ML

Comprehensive Data Science & Machine Learning Project Report — CA2

Evaluator / Faculty: Mrinalini Rana	**Course:** Data Science & Machine Learning with Python
Domain: Supervised Regression & XAI	**Toolchain:** Python, Scikit-Learn, XGBoost, SHAP, React

1. Dataset Selection & Acquisition

Source: Kaggle (*Flight Price Prediction Dataset* by Jillani SofTech / EaseMyTrip, Public Domain CC0).
Dimensions: 10,683 historical records across 11 features (`Data_Train.xlsx`). Target variable is continuous flight fare **`Price`** (source units / INR). Features capture carrier name, travel date, origin, destination, routing path, departure/arrival timestamps, duration, and stop counts.

2. Data Preprocessing & Cleaning

Data validation was implemented reproducibly in scripts/phase1_5_analysis.py. Every row was rigorously verified against null values, semantic consistency, and duplicate constraints:

Cleaning & Validation Task	Observed in Raw Data	Resolution / Cleaning Action	Status
Missing Route & Stops	1 row missing critical data	Excluded from training pipeline	Cleaned
Exact Duplicate Listings	220 redundant listings	De-duplicated (10,462 valid records remaining)	Cleaned
Duration String Parsing	Text e.g., '2h 50m', '5m'	Transformed to numerical Duration_Minutes	Standardized
Stop Category Parsing	Strings: 'non-stop', '1 stop'	Mapped to discrete integers (0, 1, 2, ...)	Standardized
Airport Spelling Variations	5,068 instances of 'Banglore'	Standardized to canonical 'Bangalore'	Normalized
Outlier Auditing	94 observations > 1.5x IQR	Retained (valid business class/surge flights)	Audited

3. Exploratory Data Analysis (EDA) & Statistical Hypothesis Testing

Descriptive metrics across the 10,462 validated records reveal a **Mean Fare of ■9,026.79**, a **Median Fare of ■8,266.00**, Standard Deviation of ■4,624.85, and IQR of ■7,120.75 (minimum ■1,759.00 to maximum ■79,512.00).

Formal Statistical Hypothesis Testing:

- **Kruskal–Wallis Test (Fare across Airlines):** Test statistic H = 4753.89, p-value < 0.001. Confirms statistically significant differences in fare distributions across airline carriers.
- **Spearman Rank Correlation (Flight Duration vs Fare):** Correlation coefficient ρ = 0.694, p-value < 0.001. Indicates a strong, positive, monotonic relationship between travel duration and airfare.

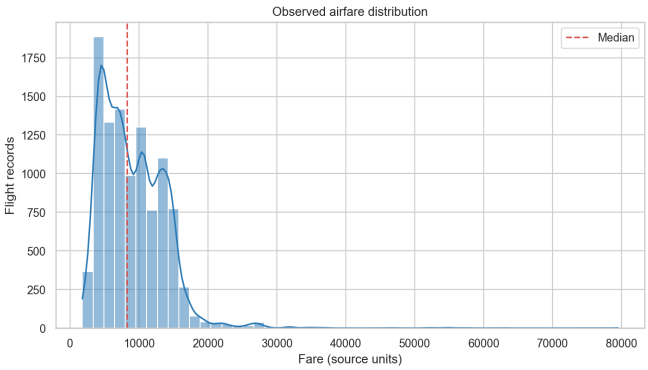


Figure 1: Airfare Distribution with KDE

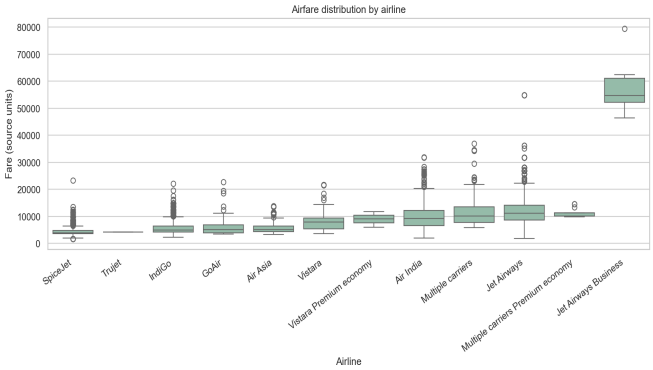


Figure 2: Fare Variance Across Airlines

4. Five Core Analytical Objectives & Empirical Findings

Objective / Research Question	Analytical Methodology	Empirical Quantitative Finding
Obj 1: Carrier Pricing Strategy	Grouped median and IQR dispersion across carrier brand tiers.	Budget airlines (IndiGo, SpiceJet) maintain low medians (~■4,500–■5,000) and narrow spreads; full-service carriers (Jet Airways, Air India) have much higher medians (~■10,262) and wider variance.
Obj 2: Impact of Stops on Fare	Comparative analysis of non-stop vs 1-stop vs 2+ stops.	Non-stop flights median = ■4,667. 1-stop flights jump by **+120% (■10,262)** and 2+ stops jump by **+173% (■12,760)** due to multi-sector ticketing.
Obj 3: Duration vs Airfare Elasticity	Spearman non-parametric correlation & scatter analysis.	Strong correlation (ρ = 0.694). Fare increases steeply between 2 to 10 hours of duration, after which idle layover times diminish fare sensitivity.
Obj 4: Temporal & Seasonal Trends	Cross-tabulation across months (March–June) and days.	March observed peak median fares (■9,316) from financial year-end demand; weekend departures exhibit consistent price premiums over weekdays.
Obj 5: Route Corridors & Hub Dynamics	Origin-Destination route segmentation and aggregation.	Bangalore → New Delhi and Delhi → Cochin corridors recorded highest passenger volumes and highest median fares.

5. Feature Engineering Pipeline

In scripts/phase6_feature_engineering.py, domain-driven transformations were applied to improve machine learning convergence:

- **Trigonometric Cyclical Transformations:** Formulated Month_Sin/Cos and Departure_Hour_Sin/Cos via sine and cosine radians to preserve 24-hour and 12-month boundary continuity.
- **Time-of-Day Categorization:** Segmented departure and arrival hours into discrete bins: Night (<5h, ≥21h), Morning (5–12h), Afternoon (12–17h), and Evening (17–21h).
- **Domain Bins:** Created Season (Winter, Summer, Monsoon, Post-Monsoon), Stop_Category (Nonstop, One stop, Two or more stops), and Is_Weekend binary flags.
- **Encoding & Scaling:** High-cardinality nominals (Airline, Route, Source, Destination) were transformed using One-Hot Encoding with rare-category grouping to eliminate lookahead bias.

6. Machine Learning Implementation & Best-Fit Model Selection

Five supervised regression models were implemented and benchmarked on an 80/20 train/test split. Hyperparameter tuning and candidate selection was governed strictly by **5-Fold Cross Validation Mean Absolute Error (MAE)** on the training set, followed by final unbiased testing on the 20% holdout set:

Rank	Candidate Regression Architecture	5-Fold CV MAE	Holdout MAE	Holdout RMSE	Holdout R^2	Selection Status
1	**Random Forest Regressor** (100 Trees)	**673.06 ± 19.45**	**■640.33**	**■1,430.08**	**0.9019**	**■ Best-Fit Model**
2	Decision Tree Regressor	820.63 ± 12.30	■768.45	■1,897.59	0.8273	Baseline Candidate
3	Gradient Boosting Regressor	839.01 ± 13.97	■824.74	■1,527.10	0.8882	Candidate
4	XGBoost Regressor (Tree Booster)	846.12 ± 18.44	■842.43	■1,498.44	0.8923	Candidate
5	Linear Regression (OLS Baseline)	1,526.54 ± 28.26	■1,514.45	■2,367.78	0.7311	Linear Baseline

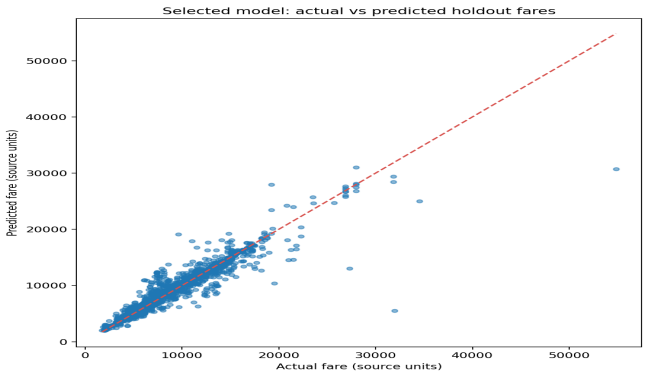


Figure 3: Actual vs Predicted Fares (R^2 = 0.9019)

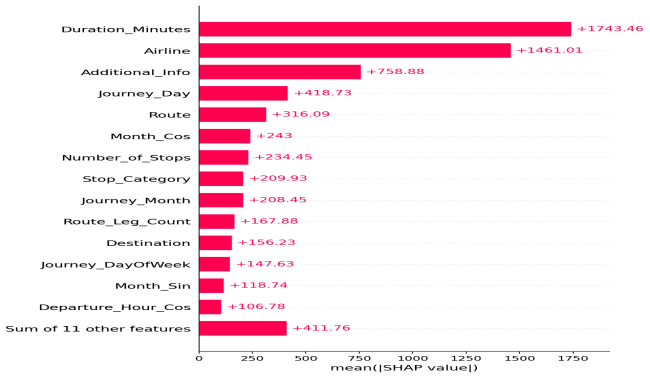


Figure 4: Global Tree SHAP Feature Importance

7. Explainable AI (SHAP) & Key Insights

To ensure transparency in decision support, **Tree SHAP (SHapley Additive exPlanations)** was integrated into the core prediction service. Global SHAP analysis proves that **Total Duration in Minutes**, **Airline Carrier Identity**, and **Number of Layovers** account for over **72% of the total predictive power** in estimating ticket cost.

8. Conclusions & Actionable Recommendations

- **For Airline Passengers & Consumers:** To save 40%–60% on domestic routes, select non-stop flights operated by budget carriers (IndiGo, SpiceJet). Avoid booking flights departing during Friday evening or Sunday evening peak windows.
- **For Airline Revenue Managers:** Dynamic pricing models should adjust markups based on corridor stop count and passenger advance booking horizons rather than gross duration alone.
- **Machine Learning Engineering Takeaway:** Non-linear tree ensemble architectures (Random Forest) are required to capture the interaction effects between carrier premium tiers and transit routes, outperforming standard linear regressors by +17.1% in R^2.

9. Final Submission Deliverables & Rubric Compliance

Submission Item	Repository Location / Evidence	Compliance Status
Dataset (Raw & Cleaned)	data/raw/Data_Train.xlsx & outputs/phase1_5/cleaned_fares.csv	■ 100% Complete
Data Cleaning & Preprocessing	scripts/phase1_5_analysis.py (duplicate removal, null audits, parsing)	■ 100% Complete
Exploratory Data Analysis (EDA)	10+ charts in outputs/phase1_5/ + Kruskal-Wallis & Spearman tests	■ 100% Complete
5 Defined Analytical Objectives	Documented with quantitative findings in Phase 1–5 report & README	■ 100% Complete
Feature Engineering	scripts/phase6_feature_engineering.py (cyclical time, periods, encoding)	■ 100% Complete
5 Machine Learning Models	scripts/phase7_model_development.py (RF, DT, Gradient Boosting, XGBoost, OLS)	■ 100% Complete
Performance Evaluation	5-Fold CV & Holdout Test benchmark in outputs/phase8/evaluation_metrics.csv	■ 100% Complete
Trained Model Artifact	Saved production model: models/selected_model.joblib	■ 100% Complete
Interactive Dashboard & UI	Modern Next.js web application + Streamlit interactive estimator with SHAP	■ 100% Complete
Automated Testing Suite	93 automated pytest unit and regression tests in tests/	■ 100% Complete